

ZEIN HAMMAD MARHABAH

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PROFESSIONAL SUMMARY

Software Engineer with 7+ years building platform-level frontend systems for multi-tenant B2B SaaS and regulated fintech, specialized in TypeScript, React, and Next.js. Deep experience in schema-driven UI engines, OpenAPI-to-typed-client codegen, and routing/data-loading architecture, and have led a frontend squad end to end. Now deliver full-stack through an AI-native workflow, directing coding agents (Claude Code over MCP) across the cycle while owning architecture, security, and code quality.

EDUCATION

STMIK Antarbangsa
Undergraduate – Computer Science (2020-2022)

WORK EXPERIENCE

Tamakun (Auto Mobility / Best Shield) | AI Fullstack Engineer (Part time)

August 2026 – Present | Riyadh

Worked on AutoConnect, AutoChat, AutoTicket, AutoSend, and a WhatsApp edge service.

Part of an AI-native team where agentic AI is built into the whole delivery cycle. Claude Code is wired to our internal engineering platform via MCP, so an agent picks up an assigned task with its full project context, works on a dedicated branch, writes the code and tests, reads real CI and production logs when debugging, and opens a PR for human review. I direct and review the agent's work, set its guardrails through rule files and reusable skills, and remain accountable for architecture, security and code quality.

AutoConnect (primary engineer)

A multi-tenant omnichannel messaging hub. Each store connects its social and messaging accounts once (WhatsApp, Messenger, Instagram, X, TikTok, Snapchat, SMS, email), and every other company app then uses one unified API with store/channel-scoped machine tokens and HMAC-signed webhooks.

- Primary engineer (~80% of commits): built it from a stripped boilerplate to a deployed platform, with a code-driven channel-capability registry so adding a platform is a descriptor plus a connector.
- Designed a connector framework and shipped 7 platform connectors, including OAuth 1.0a implemented from the RFC (X) and multi-number WhatsApp routing.
- Built an AES-256-GCM credentials vault with a distributed-lock token-refresh worker, an OAuth2 machine-token platform with per-store/per-channel grants, and an idempotent unified send API over a transactional outbox.
- Ran the pre-launch security audit (privilege-escalation, SSRF, and TLS-downgrade fixes), implemented entity-scoped CASL RBAC and passwordless OTP login, and delivered the Arabic/RTL-first dashboard.

AutoTicket

A customer-support ticketing system for Best Shield, with a customer-facing surface and a staff console.

- Owned the escalation and supervisor workflows: split the supervisor role into tickets vs escalation supervisors, added cross-department routing rules, and fixed inbound-reply and note-routing bugs.

AutoChat

A unified customer-service inbox (WhatsApp, Instagram, Messenger, TikTok, email) with tickets, an AI bot layer, CSAT and analytics.

- Made AutoChat the first production consumer of AutoConnect via a per-channel routing flag, and scoped performance KPIs per team to stop cross-team skew.

AutoSend

A multi-channel marketing / bulk-send platform and CRM for Saudi e-commerce stores.

- Made campaign send-flows runnable with step validation, and fixed channel-secret masking to obey the read contract so ciphertext no longer leaked back to the client.

WhatsApp edge service

- Added Meta X-Hub-Signature-256 verification to the legacy PHP webhook edge, shipped behind a flag for a safe rollout. (PHP)

Tech stack: TypeScript, NestJS, Next.js (App Router), PostgreSQL, TypeORM, Redis, RabbitMQ, Docker, CASL, OAuth 2.0/1.0a, Tailwind, shadcn/ui, next-intl, Playwright, GitHub Actions

PT Ajaib Sekuritas Asia | Senior Web Engineer (Contract via Accord Innovations)

March 2026 – June 2026 | Jakarta

- Delivered 5+ operator-facing features across Primary Bonds, US Stocks, and Mutual Funds domains (SID Generation, Alpaca Money Movement, MF Processing Flow), reducing manual ops effort on regulated financial workflows.
- Built customer-facing US Stocks components inside Ajaib's mobile app webview, shipping UI across the web-in-native boundary in the live product.
- Cut the ops account landing page load time by refactoring the render strategy and tuning the Next.js configuration.

Tech stack: TypeScript, React, Next.js, Jenkins

RedSquare Software MY | AI Fullstack Engineer Freelance

February 2026 – July 2026 | Kuala Lumpur

Worked on Trader Compass and ForteFi projects

Trader Compass

A trading journal and analytics platform for forex and crypto traders. Traders connect their accounts and get automatic trade syncing, performance reports, an AI trading coach, an economic calendar, and community features: gurus (mentor traders with their own member dashboards), competitions and leaderboards. Subscriptions are billed through Stripe.

- Redesigned the trading performance report, frontend and backend: balance and drawdown charts, trade heatmap, trade history, absolute gain, max drawdown and pips calculations

- Built public shareable reports with a blurred teaser gate for visitors who aren't logged in
- Made the app mobile responsive and turned it into an installable PWA (Progressive Web App)
- Added Chinese localization, a referral page, a list-view economic calendar, and a Polymarket price-history integration

Tech stack: React, TypeScript, Vite, TanStack Query, Zustand, Tailwind CSS, shadcn/ui, Recharts, TradingView Lightweight Charts, NestJS, TypeORM, PostgreSQL, Socket.IO, OpenAI API, Stripe, AWS (S3, CodeDeploy)

ForteFi

A digital investment and fund management platform for the Malaysian market. Individual and corporate clients go through KYC onboarding (NRIC, SSM company data, World-Check screening) and a risk assessment, then invest in recommended fund portfolios. An admin portal handles order management, NAV reconciliation, portfolio rebalancing, virtual accounts, fees and audit trails.

- Build the full landing page at lightning speed by connecting Figma, Framer and Claude code via MCP
- Improved automated KYC compliance documents: updated the World-Check screening section of the board resolution document the backend generates.
- Improved corporate client onboarding: pre-filled beneficial owners and controlling persons from Malaysia's company registry (SSM) data, so users no longer enter them by hand.
- Improved onboarding validation and UX: stricter input format rules, a smoother NRIC input for individuals, and a cap on nature-of-business entries enforced in the database schema (Prisma migration).
- Improved how clients see and copy their virtual bank account details on the investment dashboard.

Tech stack: Framer, React, TypeScript, Vite, TanStack Query, Zustand, Tailwind CSS, shadcn/ui, NestJS, Fastify, Prisma, PostgreSQL, Redis, Puppeteer, AWS S3, Railway

Cut and Go Grand Mall Batam | Fullstack Engineer Freelance

August 2025 – November 2025, maintained since | Batam

An IoT self-service barbershop system that I took from early exploration to live production, where it has processed thousands of paid orders. It was built so the client could leave a franchise platform and stop paying its fees and revenue share. It also closes the gaps staff could use to manipulate orders and cash. Customers order and pay at a kiosk. The system controls the power at every station through a Raspberry Pi: the clippers, dryer and steamer turn on only for a recorded order, run only as long as the paid service, and turn off automatically. Every power event is logged, so a hairstylist can't serve customers off the books or keep the cash.

- Built the platform from scratch: monorepo architecture, authentication, role-based access, and dashboards for admin, supervisor and hairstylist.
- Integrated Raspberry Pi station controllers (IoT) so equipment power follows each order and its service time, with power-usage logs for auditing.
- Built fraud-prevention controls: void-request approvals, attendance with selfie verification, and payment rules enforced on the server for cash and QRIS (Midtrans).
- Built a reporting suite for sales, transactions, orders and attendance, with Excel export, so owners can see revenue and each hairstylist's performance.
- Delivered a kiosk and dashboard in Indonesian and English that work on mobile.

Tech stack: TypeScript, React, Vite, TanStack Router/Query/Form/Table, oRPC, Zod, Hono, Better Auth, Drizzle ORM, PostgreSQL, Tailwind CSS, shadcn/ui, Recharts, Midtrans (QRIS), Raspberry Pi (IoT), Docker, Railway, Turborepo

Hyperscal | Frontend Team Lead (Contract via TOGI)

July 2025 – March 2026 | Jakarta

A multi-tenant B2B SaaS business-support platform. Organizations model their structure as sub-tenant entities with physical sites and nestable teams, and run case management, knowledge management, workforce/shift management, email-to-case, multi-channel messaging, and RBAC on top.

- Led the frontend architecture: the Remix and React Router v7 (SSR) migration with RFC-driven flat-routes restructuring, schema-first DataView/form adoption, the OpenAPI typed-client codegen strategy, and the RBAC model (a single server-side permission prefetch cached in React Query, with synchronous access checks).

- Built the in-house schema-first DataView engine: it renders Table/Tree/List/Expandable views from a backend-served schema with sorting, pagination, persisted column settings, bulk actions, and custom cell renders, so the team ships dozens of consistent CRUD screens without manual column definitions.
- Built a config-driven form generator (~19 field types with array/group nesting and a grid layout) and a schema-independent filter/query-criteria builder that composes rules into TanStack Query keys.
- Set up the auto-generated typed API SDK from OpenAPI specs (kubb): ~5,000 generated files where types flow spec → DTO → client → Query hooks, giving compile-time safety from the network boundary into components.
- Built the organization and entity-management domain end to end (teams, sites, contacts, policies, permissions, email-to-case, messaging templates, shift/workforce config), plus auth flows (login, MFA/OTP, recovery, invitations) and the activity-log system.
- Established the Playwright end-to-end testing strategy (BDD-style Page Object Model, role-based auth setup, multi-worker isolation, CI retries, trace capture), and managed the team's workload reporting to the CTO, delivering the platform ahead of the parent-company deal renewal.

Tech stack: TypeScript, Remix, React Router v7, TanStack Query/Table, kubb, Playwright, AWS, Turborepo, Tailwind, Storybook

SERA (Serasi Autoraya) by Astra International | Frontend Engineer

November 2023 – July 2025 | Jakarta

- Built multiple car- and bus-rental web applications from scratch in Next.js under the TRAC brand, delivering end-to-end booking and payment flows.
- Architected a monorepo spanning car rental, bus rental, airport transfer, and attractions, with a shared component and utility library for UI consistency.
- Integrated TRAC services into Bank Mandiri's Sukha super-app, implementing RSA encryption to secure the payment payload and API handshake.

Tech stack: TypeScript, Next.js, Redux, Nx monorepo, Jenkins

Tokenomy by Indodax | Frontend Engineer

April 2021 – November 2023 | Jakarta

- Developed and maintained the Tokenomy dashboard: investment submission (fixed deposits, staking, dual currency), crypto loans, exchange transactions, and TradingView/WebSocket contract markets.
- Improved GTmetrix and PageSpeed scores by optimizing API calls and refactoring render cycles, reducing LCP and CLS.
- Built an in-house design system and component library with Storybook, and handled Sentry/Jenkins monitoring and code review.

Tech stack: Typescript, React, Redux, Storybook, WebSocket, Sentry, Jenkins, React Testing Library, Tailwind

CODEX powered by Telkom Indonesia | Frontend Engineer

June 2020 – April 2021 | Jakarta

- Developed and maintained the Bonum POS dashboard (stock management, cashier, and staff management) for SMEs.
- Built a CRM dashboard for internal use and a CMS for the blog pages, with unit tests and web performance improvements.

Tech stack: React, JavaScript, Jenkins, Jest

PT Sama Satu Jalan (Tebengan) | Frontend Engineer

November 2019 – May 2020 | South Tangerang

- Developed and maintained the internal tools dashboard for a ride-sharing app.
- Integrated and maintained user and driver tracking history with Leaflet.

Tech stack: Reactjs, Ant Design, Leaflet, Bitbucket

Garena Indonesia | Web Developer (Internship)

August 2019 – November 2019 | Jakarta

- Built and maintained embedded in-game event pages for Free Fire, Arena of Valor, GKart, and Call of Duty Mobile.
- Built tournament and voting event web apps and game landing pages.

Tech stack: Reactjs, JavaScript, HTML, CSS

SKILLS

Hard Skills: JavaScript/TypeScript, React, Next.js, React Router v7, Remix, Redux, Zustand, Zod, TanStack Query, HTML/CSS, SCSS, Tailwind, Storybook, Material UI, Ant Design, shadcn/ui, REST/GraphQL, WebSocket, Jest, React Testing Library, Playwright, Vitest, Nx, Turborepo, Node.js/Express, Nest.js, PostgreSQL, MongoDB, kubb OpenAPI codegen, CI/CD (Jenkins, GitHub Actions), Sentry, Vercel/Netlify/Railway, schema-driven and event-driven UI architecture, AI-assisted coding (Claude Code/MCP, Cursor)

Soft Skills: Technical Leadership, Frontend Architecture, Problem Solving, Cross-functional Collaboration, Agile/Scrum

Languages: Indonesian (Native), English (B2)

CERTIFICATIONS

- Fullstack Javascript - [Gdrive](#)
- Indonesian Scrum Master - [Gdrive](#)

TRAINING / WORKSHOP

- Impact Byte - Full Stack JavaScript Bootcamp - 2019